Due by2359 hours on 20240618

Math **163** Name (PLEASE PRINT) Captain Yellowbeard

Exam **2B** **2024** Fall/Spring/**Summer I** /Summer II Score _____

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	2	4								6	
1b	2	2	2							6	
1c											
2a	6									6	
2b	4	2								6	
2c											
3a	1	1	1	1	1	1				6	
3b	6									6	
3c											
								Total		36	

Use these instructions for the remainder of the exam.

Your J-number $JN := \underline{\quad 00407205 \quad} := (J_k | k \in 1..8)$

Hence

$J_1 = 0$ $J_2 = 0$ $J_3 = 4$ $J_4 = 0$

$J_5 = 7$ $J_6 = 2$ $J_7 = 0$ $J_8 = 5$

Generate your connective list using the following Table, as illustrated in the example below:

Table

0	1	2	3	4	5	6	7	8	9
$\vee$	$\wedge$	$\longrightarrow$	$\longleftrightarrow$	$\vee$	$\wedge$	$\longrightarrow$	$\longleftrightarrow$	$\vee$	$\wedge$

J-number	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8
My J-values	0	0	4	0	7	2	0	5
My C-values	$\vee$	$\vee$	$\vee$	$\vee$	$\longleftrightarrow$	$\longrightarrow$	$\vee$	$\wedge$
My Connective list	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8

My Connective List

$C_1 = \vee$ $C_2 = \vee$ $C_3 = \vee$ $C_4 = \vee$

$C_5 = \longleftrightarrow$ $C_6 = \longrightarrow$ $C_7 = \vee$ $C_8 = \wedge$

Example

The following table is worked out for the J-Number: J00230711

It illustrates how one obtains a list of connectives from one's J number.

The algorithm appears on the following page.

Truth-Value Number

Your J-number = 00407205

Keep every 0 in your J-number and replace every non-zero number in your J-number by 1 to get

Your (truth-value) **T-number**:= 00101101

Define $\forall k \in 1..8$,

T_k ≔ the k[th] entry in your **truth-value J-number** counting from the left. Therefore,

$T_1 \coloneqq 0$

$T_2 \coloneqq 0$

$T_3 \coloneqq 1$

$T_4 \coloneqq 0$

$T_5 \coloneqq 1$

$T_6 \coloneqq 1$

$T_7 \coloneqq 0$

$T_8 \coloneqq 1$

6 (1a) Find the truth-tables for after **filling in the blanks** appropriately:

2 (i) $\varphi_{1a(i)} \coloneqq \left(p\ C_1(\neg q) \right) C_2 \left((\neg p) C_3 q \right) = \left(p\ \vee\ (\neg q) \right) \vee \left((\neg p)\ \vee\ q \right)$

p	q	$\neg p$	$\neg q$	$p\ C_1(\neg q)$	$(\neg p) C_3 q$	$\left(p\ C_1(\neg q)\right) C_2\left((\neg p) C_3 q\right)$
p	q	$\neg p$	$\neg q$	$p\ \vee (\neg q)$	$(\neg p)\ \vee\ q$	$\left(p \vee (\neg q)\right)\ \vee\ \left((\neg p) \vee q\right)$
0	0	1	1	1	1	1
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	1	0	0	1	1	1

4 (ii) $\left(p C_4 q \right) C_5 \left(q C_6 r \right) = \left(p \vee q \right) \leftrightarrow \left(q \rightarrow r \right)$

p	q	r	pC_4q	qC_6r	$(pC_4q)C_5(qC_6r)$
p	q	r	$p\ \vee\ q$	$q \rightarrow r$	$(p \vee q) \leftrightarrow (q \rightarrow r)$
0	0	0	0	1	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	1	0	0
1	1	1	1	1	1

6 (1b) Find truth-tables for the following after **filling in the blanks** appropriately:

2 (i) $LS \coloneqq \left(\left(pC_7q\right)C_8r\right) = \left(\left(p \ \vee \ q\right) \wedge \ r\right)$,

2 (ii) $RS \coloneqq \left(\left(pC_8r\right) \ C_7 \ \left(qC_8r\right)\right) = \left(\left(p \ \wedge \ r\right) \ \vee \ \left(q \ \wedge \ r\right)\right)$, and

2 (iii) decide, if $LS \coloneqq \left(\left(pC_7q\right)C_8r\right) \Longleftrightarrow \left(\left(pC_8r\right) \ C_7 \ \left(qC_8r\right)\right) =: RS$

p	q	r	$p \ C_7 q$	$(pC_7q)C_8r$	pC_8r	qC_8r	$(pC_8r)C_7(qC_8r)$	$LS \Longleftrightarrow RS$
p	q	r	$p \vee q$	$(p \vee q) \wedge r$	$p \wedge r$	$q \wedge r$	$(p \ \wedge \ r) \ \vee (q \ \wedge \ r)$	$LS \Longleftrightarrow RS$
0	0	0	0	0	0	0	0	|
0	0	1	0	0	0	0	0	|
0	1	0	|	0	0	0	0	|
0	1	1	|	|	0	|	|	|
1	0	0	|	0	0	0	0	|
1	0	1	|	|	|	0	|	|
1	1	0	|	0	0	0	0	|
1	1	1	|	|	|	|	|	|

$LS \coloneqq \left(\left(p \vee q\right) \wedge r\right) \Longleftrightarrow \left(\left(p \wedge r\right) \ \vee \ \left(q \wedge r\right)\right) =: RS$ (Y) N (Pf) W

See the boxed ones

6 (2a) Find $\varphi(p,q,r)$ in terms of p,q,r, and connectives, if $\varphi(p,q,r)$ is to have the following truth table:

p	q	r	$\varphi(p,q,r)$	φ_1	φ_2	φ_3	φ_4	φ_5	φ_6	φ_7	φ_8
0	0	0	$T_1=0$	0	0	0	0	0	0	0	0
0	0	1	$T_2=0$	0	0	0	0	0	0	0	0
0	1	0	$T_3=1$	0	0	0	0	0	1	0	0
0	1	1	$T_4=1$	0	0	0	0	1	0	0	0
1	0	0	$T_5=1$	0	0	0	1	0	0	0	0
1	0	1	$T_6=1$	0	0	1	0	0	0	0	0
1	1	0	$T_7=1$	0	1	0	0	0	0	0	0
1	1	1	$T_8=1$	1	0	0	0	0	0	0	0

$\varphi(p,q,r)$

$$
\begin{aligned}
= \;& (\varphi_1(p,q,r)) \\
& \vee (\varphi_2(p,q,r)) \\
& \vee (\varphi_3(p,q,r)) \\
& \vee (\varphi_4(p,q,r)) \\
& \vee (\varphi_5(p,q,r)) \\
& \vee (\varphi_6(p,q,r))
\end{aligned}
\qquad
\begin{aligned}
= \;& (p \wedge q \wedge r) \\
& \vee (p \wedge q \wedge (\neg r)) \\
& \vee (p \wedge (\neg q) \wedge r) \\
& \vee (p \wedge (\neg q) \wedge (\neg r)) \\
& \vee ((\neg p) \wedge q \wedge r) \\
& \vee ((\neg p) \wedge q \wedge (\neg r))
\end{aligned}
$$

6 (2b) **Show work on pages that follow**, properly numbered, to find, **using the algebra of propositions**, and the **identities expressing $p \longrightarrow q$ and $p \longleftrightarrow q$ in terms of the connectives: ¬, ∨, and ∧** to determine the equivalent expressions for $\psi(p,q)$ as a disjunction of (conjunction of literals) (called DNF) and as a conjunction of (disjunction of literals) (called CNF). Cross out the disjuncts, respectively conjuncts, from the disjunctions and conjunctions provided in the lines marked DNF and CNF respectively so that equality holds.

4 (i)

$$\psi(p,q)$$

$$:= \quad \neg\Big(\Big(p\ C_1(\neg q)\Big)\ C_2\Big((\neg p) C_3 q\Big)\Big)$$

$$= \quad \neg\Big(\Big(p\ \vee(\neg q)\Big) \vee \Big((\neg p)\ \vee\ q\Big)\Big)$$

$$= \quad \cancel{\Big(p \wedge q\Big) \vee \Big(p \wedge (\neg q)\Big) \vee \Big((\neg p) \wedge q\Big) \vee \Big((\neg p) \wedge (\neg q)\Big)\ (\text{DNF})}$$

$$= \quad \cancel{\Big(p \vee q\Big) \wedge \Big(p \vee (\neg q)\Big) \wedge \Big((\neg p) \vee q\Big) \wedge \Big((\neg p) \vee (\neg q)\Big)\ (\text{CNF})}$$

2 (ii) Fill in the column for $\psi(p,q)$, **using ONLY the columns of conjuncts (respectively disjuncts) as needed**, on the basis of your answers to part (2b)(i). Fill in the last column from (1a)(i). **Wherever $\psi(p,q)$ has 0, $\varphi_{1a(i)}$ should have 1 and vice versa.** If this is not the case, something is wrong and **0 points will be awarded for both (1a)(i) and (2b)**.

p	q	$\neg p$	$\neg q$	$p \wedge q$	$p \wedge (\neg q)$	$(\neg p) \wedge q$	$p \wedge (\neg q)$	$(\neg p) \wedge (\neg q)$	$\psi(p,q)$	$\varphi_{1a(i)}$
0	0	1	1	0	0	0	0	1	0	1
0	1	1	0	0	0	1	0	0	0	1
1	0	0	1	0	1	0	1	0	0	1
1	1	0	0	1	0	0	0	0	0	1

p	q	$\neg p$	$\neg q$	$p \vee q$	$p \vee (\neg q)$	$(\neg p) \vee q$	$p \vee (\neg q)$	$(\neg p) \vee (\neg q)$	$\psi(p,q)$	$\varphi_{1a(ii)}$
0	0	1	1	0	1	1	1	1	0	1
0	1	1	0	1	0	1	0	1	0	1
1	0	0	1	1	1	0	1	1	0	1
1	1	0	0	1	1	1	1	0	0	1

$$\neg((p \vee (\neg q)) \vee ((\neg p) \vee q))$$

$$= ((\neg(p \vee (\neg q))) \wedge (\neg((\neg p) \vee q))) \quad (\because (\neg(p \vee q)) = (\neg p) \wedge (\neg q))$$

$$= (((\neg p) \wedge (\neg(\neg q))) \wedge ((\neg(\neg p)) \wedge (\neg q))) \quad (\because (\neg(p \vee q)) = (\neg p) \wedge (\neg q))$$

$$= (((\neg p) \wedge q) \wedge (p \wedge (\neg q))) \quad (\because (\neg(\neg p) = p)$$

$$= ((p \wedge (\neg p)) \wedge (q \wedge (\neg q))) \quad (\because (p \wedge q) \wedge r = p \wedge (q \wedge r))$$

$$= ((\bot) \wedge (\bot)) \quad (\because p \wedge (\neg p) = \bot)$$

$$= \bot \quad (\because p \wedge p = p)$$

$$= \bot \ (\text{DNF})$$

$$= \bot \ (\text{CNF})$$

6 (3a) Show work on **pages that follow**, properly numbered, to decide if the following proposition, denoted as indicated is a tautology, contradiction, or contingency, and **circle an appropriate answer**.

1 (i) $\alpha \coloneqq \Big((\top\, C_1 p) C_3\, \bot \Big) = \Big((\top \vee\, p) \vee \bot \Big)$

(Tlg(α)) Cdn(α) Cng(α)

1 (ii) $\beta \coloneqq \Big(\top\, C_2 (p C_4\, \bot) \Big) = \Big(\top \vee (p \vee \bot) \Big)$

(Tlg(β)) Cdn(β) Cng(β)

1 (iii) $\gamma \coloneqq \Big((\bot\, C_3 p) C_5\, \top \Big) = \Big((\bot \vee\, p) \leftrightarrow \top \Big)$

Tlg(γ) Cdn(γ) (Cng(γ))

1 (iv) $\delta \coloneqq \Big(\bot\, C_4 (p C_6\, \top) \Big) = \Big(\bot \vee (p \rightarrow \top) \Big)$

(Tlg(δ)) Cdn(δ) Cng(δ)

1 (v) $\zeta \coloneqq \Big(\top\, C_5 (p C_7\, \bot) \Big) = \Big(\top \leftrightarrow (p \vee \bot) \Big)$

Tlg(ζ) Cdn(ζ) (Cng(ζ))

1 (vi) $\eta \coloneqq \Big((\top\, C_6 p) C_8\, \bot \Big) = \Big((\top \rightarrow p) \wedge \bot \Big)$

Tlg(η) (Cdn(η)) Cng(η)

3a.i

$\alpha := (\top \vee p) \vee \bot$
$= \top \vee \bot \quad (\because \top \vee p = \top)$
$= \top$

$\mathrm{Tlg}((\top \vee p) \vee \bot)$

3a.ii

$\beta := \top \vee (p \vee \bot)$
$= \top \vee p \quad (\because p \vee \bot = p)$
$= \top$

$\mathrm{Tlg}(\top \vee (p \vee \bot))$

3a.iii

$\gamma := (\bot \vee p) \leftrightarrow \top$
$= p \leftrightarrow \top \quad (\because \bot \vee p = p)$
$= p$

$\mathrm{Cng}((\bot \vee p) \leftrightarrow \top)$

3a.iv

$\delta := \bot \vee (p \to \top)$
$= \bot \vee \top \quad (\because p \to \top = \top)$
$= \top$

$\mathrm{Tlg}(\bot \vee (p \to \top))$

3a.v

$\zeta := \top \leftrightarrow (p \vee \bot)$
$= \top \leftrightarrow p \quad (\because p \vee \bot = p)$
$= p$

$\mathrm{Cng}(\top \leftrightarrow (p \vee \bot))$

3a.vi

$\eta := (\top \leftrightarrow p) \wedge \bot$
$= p \wedge \bot \quad (\because \top \leftrightarrow p = p)$
$= \bot$

$\mathrm{Cdn}((\top \leftrightarrow p) \wedge \bot)$

6 (3b) Show work on pages that follow, properly numbered, to decide, **using axioms and rules of inference**, if the following inference is valid and circle the appropriate choice. If (2c) has been submitted, then the answer to (3b) must agree with the answer of (2c); otherwise, 0 points on both.

$$Vld \quad \left(\frac{\left(pC_3q\right)\rightarrow\left(p\right) \quad \left(pC_3q\right)\rightarrow q}{\left(pC_3q\right)\rightarrow\left(pC_5q\right)} \right) \qquad (Y) \quad N \quad (Pf) \quad W$$

My problem:

$$Vld \quad \left(\frac{\left(p \vee q\right)\rightarrow\left(p\right) \quad \left(p \vee q\right)\rightarrow q}{\left(p \vee q\right)\rightarrow\left(p\leftrightarrow q\right)} \right) \qquad (Y) \quad N$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\mathrm{Vld}\left(\frac{r}{r}\right)}{\mathrm{Vld}\left(\frac{(r\to p)\wedge(r\to q)}{(r\to p)\wedge(r\to q)}\right)}\left\langle r\leftarrow((r\to p)\wedge(r\to q))\right\rangle}{\mathrm{Vld}\left(\frac{(r\to p)\wedge(r\to q)}{r\to(p\wedge q)}\right)}}{\mathrm{Vld}\left(\frac{((p\vee q)\to p)\wedge((p\vee q)\to q)}{(p\vee q)\to(p\wedge q)}\right)}\left\langle r\leftarrow((p\vee q))\right\rangle}{\mathrm{Vld}\left(\frac{((p\vee q)\to p)\wedge((p\vee q)\to q)}{(p\vee q)\to((p\wedge q)\vee(\neg p\wedge\neg q))}\right)}(\because \top=\top\vee p)}{\mathrm{Vld}\left(\frac{((p\vee q)\to p)\wedge((p\vee q)\to q)}{(p\vee q)\to(p\leftrightarrow q)}\right)}(\because((p\wedge q)\vee(\neg p\wedge\neg q))\equiv(p\leftrightarrow q))}{\mathrm{Vld}\left(\frac{((p\vee q)\to p)\wedge((p\vee q)\to q)}{(p\vee q)\to(p\leftrightarrow q)}\right)}$$